

Optional Office Hours Part II Notes

II.0 Errata

II.1 Naive Set Theory and the Metalanguage

Note how we use sets to discuss languages, and their structures, and when we prove results pertaining to properties of certain logical systems in languages (for example the Completeness Theorem), we're formulating these sets and creating these structures using what we call the "metalanguage", which more times than not in formal mathematics is the language of sets, i.e., the language consisting of both the equality symbol $=$, and the binary relation of elementhood $\in$ (recall **Example 7.2.1 part 1** in the [Week 7 Office Hour Notes](#)), i.e. the language of sets $\mathcal{L}_{\text{Set}}$ is the language defined as so:

$$\mathcal{L}_{\text{Set}} := \{ =, \in \}.$$

The question is, to create a robust theory of sets, which axioms do we use?

Before the 20th century, scholars (such as Cantor) utilized what we call "Naive Set Theory" (and it's the name of a textbook), which goes by the general philosophy that anything you can make a logical statement about, we can talk about as a set.

Axiom II.2.1. Axiom of Extensionality:

$$\forall x \forall y [\forall z (z \in x \leftrightarrow z \in y) \rightarrow x = y].$$

Axiom II.2.2. Axiom (schema) of Unrestricted Comprehension:

For every well-formed formula φ with free variables $w_1, \dots, w_n, z$ (with y is not free in φ), we have

$$\forall w_1 \dots \forall w_n \exists y \forall z [z \in y \leftrightarrow \varphi(w_1, \dots, w_n, z)], \quad (II.1)$$

In more common set notation, this axiom states that given a wff φ , there exists the set:

$$\{z : \forall w_1 \dots \forall w_n \varphi(w_1, \dots, w_n, z)\},$$

which is the bracket notation commonly used to be the set that exists that satisfies (II.1).

Important Note: We often call the "Axiom of Unrestricted Comprehension" an "axiom scheme" since it's not just one axiom, but rather an axiom indexed by the wff φ , so the theory $\Sigma_{\text{Naive Set}}$ consists of the following axioms:

$$\Sigma_{\text{Naive Set}} := \left\{ \forall x \forall y [\forall z (z \in x \leftrightarrow z \in y) \rightarrow x = y] \right\} \\ \cup \left\{ \forall w_1 \dots \forall w_n \exists y \forall z [z \in y \leftrightarrow \varphi(w_1, \dots, w_n, z)] : \varphi \text{ is a wff} \right\}.$$

This axiom scheme allows us to define the various operations that we think of when we think of Set Theory. For example, we can define the following sets/relations:

$$A \subset B \iff \forall x (x \in A \rightarrow x \in B)$$

Note that moving forward, we'll write shorthand notation in this language as follows:

$$\forall(y \in x)\varphi := \forall y (y \in x \rightarrow \varphi), \\ \exists(y \in x)\varphi := \exists y (y \in x \wedge \varphi), \\ \exists! y \varphi := \exists y \left(\varphi \wedge \forall z (\varphi_z^y \rightarrow z = y) \right), \text{ (note that this defines "unique existence", i.e. existence} \\ \text{of a UNIQUE set that satisfies this property)}$$

$$\emptyset := \{x : \neg x = x\},$$

since no element would be in that set. Moreover, given sets A and B , we define

$$A \cup B := \{x : x \in A \vee x \in B\}, \\ A \cap B := \{x : x \in A \wedge x \in B\}, \\ A \setminus B := \{x : x \in A \wedge \neg x \in B\}.$$

Moreover, for any finite collection $x_1, \dots, x_n$ of sets, we find the finite collection $\{x_1, \dots, x_n\}$ to be defined as the set

$$\{x_1, \dots, x_n\} := \{x : x = x_1 \vee x = x_2 \vee \dots \vee x = x_n\}.$$

Even though this bracket notation is unordered, we can still define the cartesian product $A \times B$, by first defining the ordered pairs using bracket notation as for any set a, b

$$(a, b) := \{\{a\}, \{a, b\}\},$$

and then defining

$$A \times B := \{x : x = (a, b) \text{ for some } a \in A \text{ and } b \in B\}.$$

Then we can define a given function $f_\varphi : A \rightarrow B$, for any wff $\varphi(x, y)$ (with free variables consisting of only x, y) such that the statement

$$\forall(x \in A) \exists! y \varphi(x, y),$$

is true (in the metalanguage), then we can define the graph of f_φ to be

$$\text{Graph}(f_\varphi) := \{(x, y) \in A \times B : \varphi(x, y)\},$$

which uniquely determines the function f_φ .

We can also define arbitrary unions and intersections as follows: Given a nonempty family $\mathcal{F} \neq \emptyset$ of sets, we have

$$\begin{aligned} \cup \mathcal{F} &:= \{x : \exists(A \in \mathcal{F})(x \in A)\}, \\ \cap \mathcal{F} &:= \{x : \forall(A \in \mathcal{F})(x \in A)\}. \end{aligned}$$

II.2 Cardinality and Transfinite Induction

Definition II.2.1. Given two sets A, B , we state that they have the **same cardinality**, and write $|A| = |B|$ if there exists a bijective function $f : A \rightarrow B$. We write $|A| \leq |B|$ if there exists an injective function $g : A \rightarrow B$, and we write $|A| < |B|$ if $|A| \leq |B|$, but not $|A| = |B|$.

Definition II.2.2. We define the **ordinal numbers** α to be the collection of sets satisfying one of the following conditions:

- (i) $\alpha = \emptyset$ (note that we set $0 := \emptyset$ and use the two symbols interchangeably)
- (ii) $\alpha = \alpha' \cup \{\alpha'\}$, for some ordinal α' .
- (iii) $\alpha = \cup \mathcal{F}$ and $\alpha \neq 0$, for some family $\mathcal{F}$ of ordinals.

We write $ON(\alpha)$ as shorthand notation for stating that α is an ordinal.

Any ordinal satisfying *property (ii)* is known as a **successor ordinal**, and any property satisfying *property (iii)* is known as the **limit ordinal**.

Defining the **successor of any set** x to be $S(x) := x \cup \{x\}$, we can then formulate the natural numbers as the following ordinals:

$$\begin{aligned} 0 &:= \emptyset \\ 1 &:= S(0) = \emptyset \cup \{\emptyset\} = \{\emptyset\} = \{0\} \\ 2 &:= S(1) = \{\emptyset, \{\emptyset\}\} = \{0, 1\} \\ 3 &:= S(2) = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\} = \{0, 1, 2\} \quad (II.2) \\ &\vdots \end{aligned}$$

so the set $\mathbb{N}$ is defined recursively to be the smallest set containing $\emptyset$ and closed under the successor

$$\mathbb{N} := \bigcap \{X : \emptyset \in X \wedge \forall (x \in X)(S(x) \in X)\}$$

We can then define arithmetic with symbols $+$, $<$ on the natural numbers (by recursion on $\mathbb{N}$) as follows:

$$\alpha + \beta := \begin{cases} \alpha & \text{if } \beta = 0, \\ S(\alpha + \beta') & \text{if } \beta = S(\beta') \text{ for some natural number } \beta', \end{cases}$$

$\alpha < \beta \iff \alpha \in \beta$ (note that this works since for every $m < n$, we have $m \in n$, since we defined $n := \{1, \dots, n-1\}$ to be the set of all predecessors)

However, we find this continues even after the set of natural numbers, and the ordinal numbers end up serving as a proper extension of the natural numbers:

Example II.2.3. We find that $\omega := \mathbb{N} \cup \{0\}$ is an ordinal, since $\omega = \bigcup_{\alpha \in \mathbb{N}} \alpha$, which satisfies *property (iii)* of **Definition II.2.2**. Additionally, we have $\omega + 1 = S(\omega)$, $\omega + 2 = S(S(\omega))$, and so on.

This ends up giving rise to transfinite induction on ordinals, which are the following two equivalent statements, Given an ordinal α :

Principle 1 of Transfinite Induction on Ordinals. if $X \subset \alpha$ is a nonempty set such that for every ordinal $\beta \in \alpha$, the statement $\forall \gamma (\gamma \in \beta \rightarrow \gamma \in X) \rightarrow \beta \in X$ is satisfied, then $X = \alpha$.

Principle 2 of Transfinite Induction on Ordinals. if $X \subset \alpha$ is set such that all the conditions

are satisfied:

- (i) (base case) $0 \in X$,
- (ii) (successor step) $\forall(\beta \in \alpha)[\beta \in X \implies S(\beta) \in X]$,
- (iii) (limit step) $\forall(\beta \in \alpha)[(\forall(\gamma \in \beta)[\gamma \in S]) \implies \beta \in S]$,

Then $X = \alpha$.

Theorem II.2.3. Every ordinal α is a well-ordered set under the relation $\in$, i.e., forms a total partially ordering such that every $X \subset \alpha$ has a minimum, i.e. an element $\beta_0 \in X$ such that there is no $\gamma \in X$ such that $\gamma \in \beta_0$ (or equivalently $\gamma = \beta_0$ or $\beta_0 \in \gamma$, for all $\gamma \in X$).

Usually when we talk about Cardinal numbers, we define them in one of two ways:

1. An equivalence class on all sets have the same cardinality (note that $|X| = |Y|$, for any two sets X, Y satisfies the properties of an equivalence relation), though note that this definition is a bit rigid to do in a rigorous way in more advanced ZFC set theory because this equivalence class unfortunately doesn't end up being a set (more on that in the next section).
2. Given an ordinal number α , the smallest ordinal number λ such that $|\lambda| = |\alpha|$ (note that λ ends up existing as a set, since the definition $\{\beta : ON(\beta) \wedge |\beta| = |\alpha|\}$ ends up being a nonempty subset of $S(\alpha)$ in ZFC set theory, allowing us to use **Theorem II.2.3** to choose the minimum element).

%EDIT THESE NOTES FROM THE NEXT THEOREM ONWARDS

Definition II.2.4. We define a **cardinal number** κ to be the smallest ordinal such that for every ordinal α such that $\alpha = \kappa \alpha \in \kappa$, we have $|\kappa| = |\alpha|$.

%WRITE MORE RIGOROUS DEFINITION IN Definition II.2.4.

II.3 Paradoxes and ZFC Set Theory

Although the idea of Naive Set theory gives us a elegant foundation of mathematics that serves as the ideal metalanguage, since it essentially allows us to formulate all of what we want in mathematics in a rigorous way, unfortunately the axioms stated in Naive Set theory back in **Section II.1** end up failing because ultimately what is allowed to be a set is "too big".

Example II.3.1. Under the axioms of Naive Set Theory, the following collections end up being sets

(i) The "set of all sets", called the universal set U defined by

$$U := \{x : x = x\}.$$

(ii) The set involved in Russel's Paradox is the "set of all sets that don't contain themselves"

$$V := \{x : \neg x \in x\}. \quad (II.3)$$

(iii) The set of all cardinal numbers, which can be defined by

$$\mathcal{K} := \left\{ \kappa : \forall \lambda \left[(ON(\lambda) \wedge |\kappa| = |\lambda|) \rightarrow (\kappa = \lambda \vee \kappa \in \lambda) \right] \right\}. \quad (II.4)$$

which is can be further elaborated as a wff in $\mathcal{L}_{\text{Set}}$.

This ends up leading to two commonly-known "paradoxes" (among I'm sure more) in Set Theory: One being *Russell's Paradox* and the other being *Cantor's Paradox*:

Paradox II.3.2. (Russell's Paradox) Let V be defined as in (II.3). We get the contradiction $V \in V$ and $V \notin V$.

Proof. In the case where $V \in V$ we find V satisfies the property $\neg x \in x$, and so it follows that $V \notin V$, contradicting the original case. However, in the case where $V \notin V$, we find that V satisfies the property $\neg x \in x$, so it follows that $V \in V$, also contradicting that case. $\square$

Paradox II.3.3. (Cantor's Paradox) Let $\mathcal{K}$ be defined as in (II.4). Then we have

$$\left| \bigcup \mathcal{K} \right| < \left| \bigcup \mathcal{K} \right|.$$

Lemma II.3.4. For any set A , we have $|A| < |\mathcal{P}(A)|$.

Proof. Given A , we find that the function $x \mapsto \{x\}$ forms an injection, hence $|A| \leq |\mathcal{P}(A)|$. But to show $|A| < |\mathcal{P}(A)|$, suppose towards contradiction there exists a bijection $f : A \rightarrow \mathcal{P}(A)$. We can define

$$B := \{x \in A : x \notin f(x)\}.$$

By surjectivity, there exists $b \in A$ such that $f(b) = B$, leaving us with two cases. In the first

case where $b \in B$, we have $b \in f(b)$, but since $b \in B$, we find $b \notin f(b)$, contradiction. However in the second case where $b \notin B$, then $b \notin f(b)$, hence we find $b \in B = f(b)$, which is also a contradiction. $\square$

Lemma II.3.5. For any cardinal κ , there exists a cardinal 2^κ such that $2^\kappa = |\mathcal{P}(\kappa)|$.

Proof. This follows from the Well-Ordering Theorem, since by that theorem, there exists a cardinal 2^κ (which we call 2^κ for certain reasons that go beyond the scope of this lecture) such that $|2^\kappa| = |\mathcal{P}(\kappa)|$.

Note that alternatively this proof can be proven using "cardinal arithmetic", which doesn't need the Axiom of Choice (see next section), that goes beyond the scope of this course. $\square$

%state that II.3.5 follows immediately by the Trichotomy law for Cardinal numbers, which we prove in Section II.4.

Lemma II.3.6. Given a set $\mathcal{A}$ of cardinals for

$$\sup \mathcal{A} := \min \left\{ \kappa : \kappa \text{ is a cardinal and } |\lambda| \leq |\kappa| \text{ for all } \lambda \in \mathcal{A} \right\},$$

$$\text{we have } |\sup \mathcal{A}| = \left| \bigcup \mathcal{A} \right|.$$

Proof. Outside the scope of this course. $\square$

Proof of Cantor's Paradox. Note that $ON\left(\bigcup \mathcal{K}\right)$, hence there exists a cardinal κ' such that

$|\kappa'| = \left| \bigcup \mathcal{K} \right|$. Then by Lemma II.3.4, we have $|\kappa'| < |\mathcal{P}(\kappa')|$. However, since

$|2^{\kappa'}| = |\mathcal{P}(\kappa')|$ and we have $2^{\kappa'} \in \mathcal{K}$, we find that $|2^{\kappa'}| \leq |\sup \mathcal{K}| = \left| \bigcup \mathcal{K} \right|$, and we conclude that

$$\left| \bigcup \mathcal{K} \right| = |\kappa'| < |\mathcal{P}(\kappa')| = |2^{\kappa'}| \leq \left| \bigcup \mathcal{K} \right|. \quad \square$$

HAD AN UNTIMELY ZOOM SHUTDOWN; TO BE CONTINUED

%Russel's Paradox

%Cantor's Paradox

%Axioms of ZFC Set Theory

Since we've shown Naive Set Theory to be contradictory, we need a different set of axioms that end up yielding a consistent form of mathematics. The "foundation of mathematics" that we use in set-based theoretical mathematics today is as known as Zermelo-Frankel-Choice Set Theory (aka ZFC set theory), where we write *ZFC* to denote the set of axioms that we state below:

%TALK ABOUT PROPER CLASSES IN A FUTURE DRAFT AND SPLIT PARADOX AND AXIOM SECTION

Axiom II.3.1. Axiom of Extensionality:

$$\forall x \forall y [\forall z (z \in x \leftrightarrow z \in y) \rightarrow x = y]$$

This equivalently can be stated as:

$$\forall x \forall y [x \subset y \wedge y \subset x \rightarrow x = y]$$

Axiom II.3.2. Axiom (schema) of Restricted Comprehension/Specification:

For every formula φ with free variables $x, w_1, \dots, w_n, z$ (with y is not free in φ), we have

$$\forall x \forall w_1 \dots \forall w_n \exists y \forall (z \in x) [z \in y \leftrightarrow \varphi(x, w_1, \dots, w_n, z)],$$

which can be equivalently stated as

$$\forall x \forall w_1 \dots \forall w_n \exists y (y \subset x \wedge \forall (z \in x) [z \in y \leftrightarrow \varphi(x, w_1, \dots, w_n, z)]).$$

Note that since there is a different axiom for every formula φ , we call this axiom an "axiom scheme".

Axiom II.3.3. Axiom of Pairing:

$$\forall x \forall y \exists z ((x \in z) \wedge (y \in z)).$$

Axiom II.3.4. Axiom of Union:

$$\forall \mathcal{F} \exists A \forall (Y \in \mathcal{F}) \forall (x \in Y) (x \in A).$$

Axiom II.3.5. Axiom of Power Set:

$$\forall x \exists y \forall z (\forall (w \in z)(w \in x) \rightarrow z \in y),$$

which can be equivalently stated more clearly as

$$\forall x \exists y \forall z (z \subset x \rightarrow z \in y).$$

Note that these axioms, used along with comprehension, state that there exists the sets that we refer to as

$$\{x, y\};$$

$$\cup \mathcal{F}, \text{ for any family of sets } \mathcal{F};$$

$$\mathcal{P}(A), \text{ for any set } A.$$

%TALK ABOUT SINGLETON SETS AND CLARIFY HOW THE AXIOM OF PAIRING IS USED REPEATEDLY

Additionally, repeatedly using the Axiom of Pairing along with the Axiom of Union gives us $\{x_1, \dots, x_n\}$ for distinct sets $x_1, \dots, x_n$, and using the axiom of comprehension (in a specific way that I'll leave to the reader) gives rise to

$$A \setminus B, \text{ for any two sets } A, B$$

$$\cap \mathcal{F}, \text{ for nonempty } \mathcal{F}.$$

Axiom II.3.6. Axiom of Infinity:

$$\exists X [\exists (e \in X) (\forall z \neg (z \in e)) \wedge \forall (x \in X) \exists (y \in X) (\forall (w \in x) (w \in y) \wedge x \in y)],$$

which can be equivalently stated more clearly as

$$\exists X [\emptyset \in X \wedge \forall (x \in X) \exists (y \in X) (x \subset y \wedge x \in y)].$$

Using the definition of ordinals that we used, we can use the axiom of infinity and axiom of comprehension to show that the set $\mathbb{N}$ exists, which gives rise to the ordinal $\omega, \omega + 1, \dots$ and so on.

This moreover gives rise to the existence of uncountable sets, since we can use the power

set axiom to formulate $\mathcal{P}(\mathbb{N})$.

%SHOW THAT THIS AXIOM JUSTIFIES THE EXISTENCE OF THE NATURAL NUMBERS
AND UNCOUNTABLE SET

Axiom II.3.7. Axiom of Foundation/Regularity:

$$\forall x \left[\exists a (a \in x) \rightarrow \exists (y \in x) (\neg \exists (z \in y) (z \in x)) \right],$$

which can equivalently be stated more clearly as

$$\forall x (x \neq \emptyset \rightarrow \exists (y \in x) (x \cap y = \emptyset)).$$

This stops *Russel's Paradox* since if there exists a set x such that $x \in x$, then $y := \{x\}$ is a set such that for every $z \in y$, we have $z = x$, therefore we have $z \in x$ and $z \in y$, making $x \cap y$ nonempty

Axiom II.3.8. Axiom (schema) of Replacement:

For every formula φ with free variables $x, y, A, w_1, \dots, w_n$ (with B not free in φ), we have

$$\begin{aligned} \forall A \forall w_1 \dots \forall w_n [& \forall (x \in A) \exists! y \varphi(x, y, A, w_1, \dots, w_n) \\ & \rightarrow \exists B \forall (x \in A) \exists (y \in B) \varphi(x, y, A, w_1, \dots, w_n)], \end{aligned}$$

which can be equivalently stated more clearly as

$$\begin{aligned} \forall A \forall w_1 \dots \forall w_n [& \forall (x \in A) \exists! y \varphi(x, y, A, w_1, \dots, w_n) \\ & \rightarrow \exists (f : A \rightarrow B) \forall (x \in A) \varphi(x, f(x), A, w_1, \dots, w_n)], \end{aligned}$$

This axiom scheme does a combination of helping us define new sets that are images of functions whose domain is itself a set (which in a standard Graduate level set theory course you'll learn is definitely needed!), but also it provides us a way to stop Cantor's Paradox, as so:

Theorem II.3.9. (Hartog's Lemma) For every set X , there exists a cardinal κ such that there is no injection exists between X and κ .

Proof. Suppose towards contradiction that a set X exists such that every cardinal κ has an injection between X and κ . Set

$$\mathcal{A} := \left\{ A \in \mathcal{P}(X) : A = f[\kappa], \text{ for some cardinal } \kappa \text{ and injection } f \right\},$$

and note by the Axiom of Restricted Comprehension and the Power Set Axiom that this is a well-defined set. Since the cardinal κ such that $A = f[\kappa]$, is uniquely determined, it follows by the axiom of replacement that

$$\mathcal{K} := \left\{ \kappa : A = f[\kappa] \text{ for some injection } f : \kappa \rightarrow A \right\},$$

is a well-defined set, which by hypothesis is a set that contains all the cardinals. Therefore, it follows by the proof of Cantor's Paradox that we have a contradiction. $\square$

%state and prove Hartog's Lemma, then prove that there is no set of cardinal numbers as a corollary

%talk about how all the axioms defined up until this point are considered the ZF axioms

This leaves us with the pretty controversial axiom known as the Axiom of Choice:

Axiom II.3.9. Axiom of Choice:

$$\forall X [\forall (x \in X) \exists a (a \in x) \implies \exists z \forall (x \in X) \exists! (y \in z) (y \in x)]$$

Can be stated more clearly as:

$$\forall X \left[\emptyset \notin X \implies \exists f : X \rightarrow \bigcup_{x \in X} x : (\forall x \in X) (f(x) \in x) \right]$$

This axiom is seen as controversial for a few "philisophical" reasons that are outside the scope of this course, but it is worth mentioning that for the reason of not always wanting to assume the axiom of choice in our metalanguage, we write ZF to talk about ONLY the Zermelo-Frankel axioms without including the axiom of choice.

%mention the question on the consistency of set theory/Skolem's "Paradox"?

II.4 Consequences of the Axiom of Choice

%existence of right inverses

%the well ordering theorem

%zorn's lemma

The following are some theorems in mathematics that we need the Axiom of Choice (**Axiom II.3.9.**) to prove.

Theorem II.4.1. (Existence of Right Inverses) Every surjective function $f : X \rightarrow Y$ has a "right inverse", i.e., a (not necessarily unique) function $g : Y \rightarrow X$ such that $f \circ g = \text{id}_Y$, i.e., the function $\text{id}_Y : Y \rightarrow Y$ defined by $\text{id}_Y(x) = x$, for all $x \in Y$.

Proof. By surjectivity, for every $y \in Y$, there exists $x \in X$ such that $f(x) = y$. We can then for each $y \in Y$, define, by the Axiom of Choice, a set $\{x_y \in X : x_y \in f^{-1}(\{y\})\}$, where the inverse image

$$f^{-1}(B) := \{a \in X : f(a) \in B\}.$$

Then we can define $g : Y \rightarrow X$ by $g(y) := x_y$, for all $y \in Y$. it's left to the reader to show that this is in fact a right inverse. $\square$

Theorem II.4.2. (Well Ordering Theorem) For every set X , there exists a "well-ordering", on that set, more specifically, a cardinal κ and a bijection $f : \kappa \rightarrow X$ that indexes all elements of X . Equivalently, we have $|X| = \kappa$, for some cardinal κ .

The proof of this theorem is outside the scope of this course, but it ends up using the Axiom of Choice, along with a combination of transfinite recursion and Hartog's Lemma. I'll link a more specific proof in a later draft.

Corollary II.4.3. (Trichotomy Law for Cardinality) For every set X and Y , we have (precisely one of) $|X| < |Y|$, $|X| = |Y|$, or $|X| > |Y|$.

Proof. Use the Well-Ordering Theorem to choose cardinals κ_X and κ_Y such that $|X| = \kappa_X$ and $|Y| = \kappa_Y$. We find since κ_X, κ_Y are ordinals, we have precisely one of $\kappa_X \in \kappa_Y$, $\kappa_X = \kappa_Y$, or $\kappa_X \ni \kappa_Y$, and our conclusion follows from the fact that since κ_X is the smallest ordinal that has the same cardinality that $\kappa_X \in \kappa_Y \implies |\kappa_X| < |\kappa_Y|$. (a fact that I'll leave to the reader using **Definition II.2.4.** $\square$

Theorem II.4.4. (Zorn's Lemma)

(i) Let $(X, \leq)$, $X \neq \emptyset$ be a poset such that every non-empty chain (i.e., a subset forming a total linear ordering) in X has an upper bound in X . Then X has at least one maximal element.

(ii) Let $\mathcal{C}$ be a collection of sets such that for every $\subset$ -chain $C \subset \mathcal{C}$, we have $\bigcup C \in \mathcal{C}$. Then there exists $X \in \mathcal{C}$ such that no set Y that properly contains X (i.e. $Y \supsetneq X$) exists.

In other words, (ii) is a special case of (i) where $\subset$ is the poset relation in place of $\leq$.

Why is it that we claim that we need the Axiom of Choice to prove all these previous aforementioned theorems. Well, that's because it's been proven that all these theorems are equivalent to the axiom of choice if we assume ZF to be our axioms instead of ZFC .

Theorem II.4.5. Assuming only the ZF axioms of Set Theory hold, then the following are equivalent to the Axiom of Choice (i.e., if we assume one of these results are true, then both the Axiom of Choice and all the others are true)

(i) Existence of Right Inverses (**Theorem II.4.1**)

(ii) The Well-Ordering Theorem (**Theorem II.4.2**)

(iii) The Trichotomy Law for Cardinality (**Theorem II.4.3**)

(iv) Zorn's Lemma (**Theorem II.4.4**)

I won't prove this theorem since it's outside the scope of this course, but I will make a reference to a proof of this theorem in a future draft.

Lastly, recall that we mentioned in the Week 14-15 Office Hour Notes that we only proved the Completeness Theorem for countable languages $\mathcal{L}$. We wouldn't be able to do the proof otherwise, since enumerating the $\mathcal{L}'$ -formulas in **Step 1** we could only do provided that the language $\mathcal{L}$ is countable.

In the case $\mathcal{L}$ is uncountable, we can fix this theorem through a few minor tweaks as follows

General Outline of the Proof of the version of the Completeness Theorem where $\mathcal{L}$ is uncountable.

Step 1. Use the Well-Ordering Theorem (Theorem II.4.2) to choose a cardinal κ such that $|\kappa| = |\mathcal{L}|$. Then let $\{c_\alpha\}_{\alpha \in \kappa}$ be a family of distinct objects that we utilize as constant symbols in the expanded language $\mathcal{L}' := \mathcal{L} \cup \{c_\alpha\}_{\alpha \in \kappa}$. Note that it can be shown (I'll leave this as an exercise to any brave reader) that

$$\left| \left\{ \varphi : \varphi \text{ is a well-formed } \mathcal{L}'\text{-formula} \right\} \right| = |\kappa|,$$

and from this we can choose a well ordering $\{\varphi_\alpha\}_{\alpha \in \kappa}$ index on the set of well-formed $\mathcal{L}'$ -formulas.

Step 2. Just like with **Step 2** in the original proof, we define a set

$\Phi := \left\{ \exists x \varphi_\alpha \rightarrow \varphi_{\alpha_{c_\alpha}}^x : n \geq 1 \right\}$ of *Henkin Witnesses* that is similar to the countable of the proof, except that the $\mathcal{L}'$ are indexed by $\alpha \in \kappa$. We then show as before that $\Gamma \cup \Phi$ remains consistent.

Step 3. Using what we call "transfinite recursion" on κ , which is a more general version of recursion on arbitrary ordinals, and hence cardinals to define Δ_α , for all $\alpha \in \kappa$ as follows:

$$\Delta_\alpha := \begin{cases} \Gamma \cup \Phi & \text{if } \alpha = 0 \\ \Delta_\beta \cup \{\varphi_\alpha\} & \text{if } \alpha = S(\beta) \text{ and } \Delta_\beta \cup \{\varphi_\alpha\} \text{ is consistent} \\ \Delta_\beta \cup \{\neg \varphi_\alpha\} & \text{if } \alpha = S(\beta) \text{ and } \Delta_\beta \cup \{\varphi_\alpha\} \text{ is not consistent} \\ \bigcup_{\gamma < \alpha} \Delta_\gamma & \text{if } \alpha \text{ is a limit ordinal} \end{cases}$$

Setting $\Delta := \bigcup_{\alpha \in \kappa} \Delta_\alpha$, we can show that Δ is maximally consistent by two straightforward proofs by contradiction (one to show that it's consistent and the other to show that it's maximal), and hence more generally we can show that it satisfies properties (i)-(iii) of the countable version of this proof.

Then we can proceed with **Steps 4-6** as shown before. $\square$

In office hours coming up this Friday, I'll show how cardinal numbers and the properties we proved here apply to the material such as the Lowenheim Skolem Theorems, Vaught's Test, etc. that we talk about in the powerpoint lectures and Section 2.6 of Enderton.

%using the well ordering theorem to generalize the proof of completeness

II.5 Ordinal and Cardinal Arithmetic

%transfinite recursion

%define ordinal addition, multiplication, and exponentiation

%define cardinal addition, multiplication, and cardinal exponentiation